

LUND
UNIVERSITY

Centre for Mathematical Sciences
Mathematics, Faculty of Science

Written Examination
Number Theory
Saturday January 19, 2019
Duration: 8.00–13.00

Solutions may of course be presented in Swedish! The sheets of the department should be used. Write your initials and at most one solution on each sheet handed in. Give careful motivations to your solutions!

1. Find the smallest integer $a > 2$ such that

$$2|a, \quad 3|a+1, \quad 4|a+2, \quad 5|a+3, \quad 6|a+4.$$

2. Evaluate the simple continued fraction $[0; \overline{1, 2, 3}]$.

3. Let $n \in \mathbb{Z}$. Verify that the prime divisors p of $2n(n+1)+1$ are of the form

$$p \equiv 1 \pmod 4.$$

4. Show that for every integer $n \geq 1$,

$$\sum_{d|n} \frac{\mu^2(d)}{\phi(d)} = \frac{n}{\phi(n)}.$$

Here μ denotes Möbius' μ-function, and ϕ Euler's ϕ-function.

5. Show that each prime $p \equiv \pm 1 \pmod 8$ can be expressed

$$p = x^2 - 2y^2$$

for some $x, y \in \mathbb{Z}$.

6. Let p be an odd prime and let $a \in \mathbb{Z}$, $\gcd(a, p) = 1$. Show that the number of solutions (x, y), where $x, y \in \{0, 1, \ldots, p-1\}$, to the congruence

$$x^2 - y^2 \equiv a \pmod p$$

is $p - 1$.